



Strathmore
UNIVERSITY

SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN INFORMATICS AND COMPUTER SCIENCE
ICS 4104 DISTRIBUTED SYSTEMS

DATE: 26th May 2025

CAT I

Time: 1 Hour

Instructions

1. This examination consists of **FOUR** questions.
2. Answer **ALL** the questions

QUESTION ONE (10 MARKS)

- a) Define the term Distributed System and briefly explain two characteristics that differentiate it from a centralized system. **(4 marks)**
- b) Explain the concept of cloud computing in the context of distributed systems. **(3 marks)**
- c) With examples, distinguish between public cloud, private cloud, and hybrid cloud deployment models. **(3 marks)**

QUESTION TWO (10 MARKS)

- a) Describe the **MapReduce programming model**, emphasizing how it handles parallel data processing. **(4 marks)**
- b) Assume you are analyzing user logs on a social media platform.
 - (i) Describe how you would apply the Map function.
 - (ii) Describe how you would apply the Reduce function. **(4 marks)**
- c) List **two benefits** of using MapReduce in a cloud computing environment. **(3 marks)**

QUESTION THREE (10 MARKS)

- a) Compare the **client-server model** and the **peer-to-peer model**, highlighting **two differences** in communication and architecture. **(4 marks)**
- b) Explain the following principles of distributed systems with examples:

(i) Concurrency

(ii) Transparency

(4 marks)

c) Why is **scalability** an important requirement in cloud-based distributed systems? **(2 marks)**

QUESTION FOUR (10 MARKS)

a) Distinguish between the following communication modes in distributed systems, providing examples:

(i) Synchronous vs. Asynchronous Communication **(4 marks)**

(ii) Remote Method Invocation (RMI) vs. Remote Procedure Call (RPC) **(6 Marks)**